University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2021
MATH-1105: Differential and Integral Calculus

Total Marks: 70 Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Define Continuity and Differentiability of a Function. 3

b) Test the Continuity and Differentiability of the following function at $x = \frac{\pi}{2}$ 6

$$f(x)=\begin{cases} 1 & when\ x < 0 \\ 1+\sin x & when\ 0 \leq x \leq \frac{\pi}{2} \\ 2+(x-(\frac{\pi}{2}))^2 & when\ x \geq \frac{\pi}{2} \end{cases}$$

c) Draw the graph of the following function $f(x) = \begin{cases} x-1\,; x>0 \\ -\frac{1}{2}\quad ; x=0 \\ x+1\,;\, x<0 \end{cases}$ 5

2. a) Explain Exponentially growth and decay with graph. 4

b) Find the Approximation of $f(x) = \sqrt{x}$ at x = 16 with graph. 4

c) State L'Hospital Rule. Evaluate (Using L'Hospital rule) 6

i. $\lim\limits_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$ ii. $\lim\limits_{x\to 0}\frac{x-\tan x}{x^3}$

3. a) Write the Walli's formula. Find the reduction formula of $\int \tan^n x dx$ 4

b) Prove that 6

$$\int_0^{\frac{\pi}{2}} \sin^p\theta \cos^q\theta d\theta = \frac{\Gamma(\frac{p+1}{2}).\Gamma(\frac{q+1}{2})}{2\Gamma(\frac{p+q+2}{2})}$$

c) Evaluate 4

$$\lim_{n\to\infty}\left[\frac{1}{n+1}+\frac{1}{n+2}+\ldots\ldots+\frac{1}{2n}\right]$$

4. a) Find maximum, minimum value and critical point of the function 7

$$f(x) = x^3 - 9x^2 + 15x - 3$$

b) If $\ln y = \tan^{-1} x$ then show that $(1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$. 7

5. a) Define gamma and beta Function. 2

b) Find the radius of the curvature of the asteroid: 5

$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ at any point.

c) Prove that 7

$$\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

6. a) Evaluate the followings (any two): 3.5x2

(i) $\int_0^1 x^3\sqrt{1+3x^4}\,dx$ (ii) $\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$ (iii) $\int_0^{\frac{\pi}{2}}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$

b) Evaluate the followings (any two): 3.5x2

(i) $\int\frac{1-\sin x}{x+\cos x}dx$ (ii) $\int \cos^5 x \sin^4 x dx$ (iii) $\int \tan^{-1} x\, dx$

7. a) Find the area included between the cycloid $x = a(\theta - \sin\theta)$ and $y = a(1-\cos\theta)$. 8

b) Find the volume formed by revolving the region bounded by the lines y = x and x = 2 about X axis. 6